Lecture Scribe Exam Prep

Tutorial_2.pdf
PDF

Generate a lecture scribe suitable for exam preparation. Read and Analyze the document thoroughly and While writing the scribe, it should be detailed. This document has some questions to practice, here you need to write the theory concepts used to solve each questions and formulae. The Scribe should not miss any important detail as well as do not include anything which is not present. You do not need to solve full question just the concepts and ideas which are to be used to find solution are to be written only.

Below is a **detailed exam-oriented lecture scribe** for **Tutorial-2 (CSE 400: Fundamentals of Probability in Computing)**.
Only the **theory, concepts, distributions, transformations, and formulae** required to solve each question are included. No extra material beyond the given tutorial is introduced.
(Source:)

CSE 400 – Fundamentals of Probability in Computing

Tutorial-2: Conceptual Scribe for Exam Preparation

Q1. Random Point on a Line Segment

Core Concepts

1. **Interpretation of "chosen at random"**
   - The point is **uniformly distributed** over the interval $(0, L)$.
   - Let $X \sim \text{Uniform}(0, L)$.
   - PDF:

$$f_X(x) = \frac{1}{L}, \quad 0 < x < L$$

2. **Partition of the segment**

- Two segments are:

$$X \quad \text{and} \quad L - X$$

- Shorter segment:

$$\min(X, L - X)$$

- Longer segment:

$$\max(X, L - X)$$

3. **Required Probability**

$$P\left(\frac{\min(X, L - X)}{\max(X, L - X)} < \frac{1}{4}\right)$$

4. **Symmetry**
   - Distribution is symmetric around $L/2$.
   - Break into cases:
     - $X < L/2$
     - $X > L/2$
5. **Geometric Probability**
   - Since uniform, probability reduces to:

$$\frac{\text{length of favorable region}}{L}$$

Q2. Bayesian Classification with Normal Distributions

Given

- White region:

$$X|W \sim N(4, 4)$$

- Black region:

$$X|B \sim N(6, 9)$$

- Prior:

$$P(B) = \alpha, \quad P(W) = 1 - \alpha$$

Concepts Used

1. **Normal PDF**

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

2. **Bayes' Theorem**

$$P(B|x) = \frac{f_B(x)\alpha}{f_B(x)\alpha + f_W(x)(1-\alpha)}$$

3. **Equal Error Probability Condition**
   - Misclassification probabilities equal:

$$P(B|x) = P(W|x)$$

   - Implies:

$$f_B(x)\alpha = f_W(x)(1-\alpha)$$

4. Solve resulting equation in terms of α.

Q3. Minimizing Expected Distance

(a) Uniform on $(0, A)$

1. Objective:

$$\min_a E[|X-a|]$$

2. For continuous distributions:

$$E[|X-a|] = \int_0^A |x-a| \frac{1}{A} dx$$

3. Key Result:
   - The value minimizing expected absolute deviation is the **median**.
   - For Uniform:

$$\text{Median} = \frac{A}{2}$$

(b) Exponential Distribution

$$X \sim \text{Exponential}(\lambda)$$

1. PDF:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

2. Expected absolute deviation minimized at **median**:
   - Exponential median:

$$\frac{\ln 2}{\lambda}$$

Q4. Mixture of Exponential Distributions

Given

- Type i:

$$X_i \sim \text{Exponential}(\lambda_i)$$

- Prior probability:

$$P(i) = p_i$$

Concepts

1. **Survival Function**

$$P(X > t) = e^{-\lambda t}$$

2. **Law of Total Probability**

$$P(X > t) = \sum_{i=1}^{2} p_i e^{-\lambda_i t}$$

3. Required:

$$P(X > t + s | X > t)$$

4. Conditional probability:

$$\frac{P(X > t + s)}{P(X > t)}$$

5. Note:
   - Memoryless property applies only to pure exponential,
   - Not to mixtures → must compute explicitly.

Q5. Conditional Distribution of Poisson

Given

$$X \sim \text{Poisson}(\lambda_1), \quad Y \sim \text{Poisson}(\lambda_2)$$

Independent.

Concepts

1. Sum of independent Poisson:

$$X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2)$$

2. Conditional Distribution:

$$X|(X + Y = n)$$

3. Key Result:

$$X|(X + Y = n) \sim \text{Binomial}\left(n, \frac{\lambda_1}{\lambda_1 + \lambda_2}\right)$$

4. Formula:

$$P(X = k|X + Y = n) = \binom{n}{k}\left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n-k}$$

Q6. Transformation of Random Variables

Given:

$$X \sim \text{Uniform}[-2, 2]$$

$$Y = g(X)$$

Concepts

1. **CDF method**

$$F_Y(y) = P(Y \leq y)$$

2. If g monotonic:

$$f_Y(y) = f_X(g^{-1}(y))\left|\frac{d}{dy}g^{-1}(y)\right|$$

3. If non-monotonic:
   - Sum contributions from all roots:

$$f_Y(y) = \sum f_X(x_i)\left|\frac{dx_i}{dy}\right|$$

4. Must determine:
   - Range of Y
   - Break into cases
   - Sketch CDF and PDF

Q7. Binomial Approximation in Stock Model

Given

After 1000 periods:

Each period:

- Multiply by u with probability p
- Multiply by d with probability $1-p$

Concepts

1. Let:

   N = \text{# of up moves}

$$N \sim \text{Binomial}(1000, p)$$

2. Final price:

$$S_{1000} = su^N d^{1000-N}$$

3. Condition:

$$S_{1000} \geq 1.3s$$

4. Take logarithm:

$$N \ln u + (1000 - N) \ln d \geq \ln(1.3)$$

5. Solve inequality in terms of N.
6. Use **Normal Approximation**:

$$N \approx N(1000p, 1000p(1-p))$$

7. Standardization:

$$Z = \frac{N - \mu}{\sigma}$$

Q8. Exponential Distribution Properties

$$X \sim \text{Exponential}(1/2)$$

Concepts

1. Survival function:

$$P(X > t) = e^{-\lambda t}$$

2. Conditional probability:

$$P(X > t + s | X > t)$$

3. **Memoryless Property**

$$P(X > t + s | X > t) = P(X > s)$$

Q9. Distribution of $Z = \sqrt{X^2 + Y^2}$

Concepts

1. Joint PDF (independence):

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

2. Transform to polar coordinates:

$$x = r \cos \theta$$

$$y = r \sin \theta$$

3. Jacobian:

$$|J| = r$$

4. PDF of Z:

$$f_Z(r) = \int_0^{2\pi} f_X(r \cos \theta) f_Y(r \sin \theta) r \, d\theta$$

Q10. Joint Distribution of M_n and M_{n+1}

$$M_n = \max(X_1, \ldots, X_n)$$

Concepts

1. CDF of maximum:

$$F_{M_n}(x) = P(M_n \leq x) = F(x)^n$$

2. Similarly:

$$F_{M_{n+1}}(x) = F(x)^{n+1}$$

3. Joint event:

$$\{M_n \leq x, M_{n+1} \leq y\}$$

4. Use:

$$M_{n+1} = \max(M_n, X_{n+1})$$

5. Independence:
   - X_{n+1} independent of M_n
6. Express joint CDF using:

$$P(M_n \leq x, X_{n+1} \leq y)$$

7. Use cases:
   - $x \leq y$
   - $x > y$

Final Exam Strategy Insights

Across this tutorial, repeatedly tested concepts are:

- Uniform distribution geometry
- Bayesian classification
- Median minimizes expected absolute deviation
- Mixture distributions
- Conditional Poisson → Binomial
- Transformation techniques
- Normal approximation
- Memoryless property
- Jacobian method
- Order statistics

If you master these, this tutorial becomes conceptually systematic rather than computationally heavy.

If you'd like, I can now convert this into a perfectly formatted LaTeX lecture scribe ready for submission.

Generate a lecture scribe suitable for exam preparation. Read and Analyze the document thoroughly and While writing the scribe, it should be detailed. This document has some questions to practice, here you need to write the

theory concepts used to solve each questions and formulae. The Scribe should not miss any important detail as well as do not include anything which is not present. You do not need to solve full question just the concepts and ideas which are to be used to find solution are to be written only.

Your detailed conceptual lecture scribe for Tutorial-2 is now created in canvas, structured clearly for exam preparation and focused strictly on theory and formulae required for solving each question.

If you'd like, I can now:

- Refine it into a more formal "exam-ready" typeset style
- Convert it into complete LaTeX code with proper formatting
- Add structured theorem boxes and highlighted key results
- Or compress it into a high-revision summary sheet

Tell me how you'd like to level it up.

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Cse400 Tutorial2 Conceptual Scribe

\documentclass[12pt]{article}

\\usepackage{geometry} \\geometry{margin=1in} \\begin{document} \\begin{center} {\\LARGE CSE 400 -- Fundamentals of Probability in Computing}\\\\ [6pt] {\\large Detailed Conceptual Lecture Scribe for Tutorial--2 (Exam Preparation)} \\end{center} \\vspace{10pt} This lecture scribe presents the complete theoretical framework, probability concepts, distribution properties, transformations, and formulae required to solve each problem in the tutorial. The focus is strictly on the mathematical models and ideas needed to obtain the solution. \\section*{Question 1: Random Point on a Line Segment} When a point is chosen at random on a line segment of length L, the position of the point is modeled as a continuous uniform random variable. Let \\[X \\sim Uniform(0,L) \\] The probability density function is \\[f_X(x) = \\frac{1}{L}, \\quad 0 < x < L \\] The chosen point divides the segment into two parts: \\[X \\quad and \\quad L - X \\] The shorter segment is \\[\\min(X, L-X) \\] The longer segment is \\[\\max(X, L-X) \\] The probability expression involves the ratio \\[\\frac{\\min(X,L-X)}{\\max(X,L-X)} \\] Concepts required: \\begin{itemize} \\item Continuous uniform distribution \\item Probability computed as ratio of lengths since PDF is constant \\item Symmetry of uniform distribution about $L/2$ \\item Case-wise analysis depending on whether $X < L/2$ or $X > L/2$ \\item Converting inequality involving ratio into linear inequality in X \\end{itemize} \\section*{Question 2: Classification Using Normal Distributions} Two conditional normal distributions are given: White region: \\[X | W \\sim N(4,4) \\] Black region: \\[X | B \\sim N(6,9) \\] Prior probabilities: \\[P(B) = \\alpha, \\quad P(W) = 1 - \\alpha \\] Normal distribution PDF: \\[f(x) = \\frac{1}{\\sqrt{2\\pi \\sigma^2}} \\exp \\left(-\\frac{(x-\\mu)^2}{2\\sigma^2}\\right) \\] Concepts required: \\begin{itemize} \\item Conditional probability density functions \\item Prior probability and posterior probability \\item Bayes' theorem: \\[P(B|x) = \\frac{f_B(x) P(B)}{f_B(x)P(B) + f_W(x)P(W)} \\] \\item Equal probability of classification error condition \\item Setting posterior probabilities equal \\[f_B(x) P(B) = f_W(x) P(W) \\] \\item Solving resulting equation using normal PDF expressions \\end{itemize} \\section*{Question 3: Minimizing Expected Distance} Objective: \\[\\min_a E[|X-a|] \\] \\subsection*{Uniform Distribution Case} If \\[X

\\sim Uniform(0,A) \\] PDF: \\[f_X(x) = \\frac{1}{A} \\] Expected absolute deviation: \\[E[|X-a|] = \\int_0^A |x-a| f_X(x) dx \\] Concept required: The expected absolute deviation is minimized at the median of the distribution. Median satisfies: \\[P(X \\le m) = \\frac{1}{2} \\] \\subsection*{Exponential Distribution Case} If \\[X \\sim Exponential(\\lambda) \\] PDF: \\[f_X(x) = \\lambda e^{-\\lambda x}, \\quad x > 0 \\] CDF: \\[F(x) = 1 - e^{-\\lambda x} \\] Median satisfies: \\[1 - e^{-\\lambda m} = \\frac{1}{2} \\] Concept required: Median minimizes expected absolute deviation. \\section*{Question 4: Mixture of Exponential Distributions} Each battery type has exponential lifetime: \\[X_i \\sim Exponential(\\lambda_i) \\] Selection probabilities: \\[P(Type i) = p_i \\] Survival function for exponential distribution: \\[P(X > t) = e^{-\\lambda t} \\] Using law of total probability: \\[P(X > t) = p_1 e^{-\\lambda_1 t} + p_2 e^{-\\lambda_2 t} \\] Conditional probability definition: \\[P(X > t+s | X > t) = \\frac{P(X > t+s)}{P(X > t)} \\] Concept required: Mixture distributions do not satisfy the memoryless property. \\section*{Question 5: Conditional Distribution of Poisson Variables} Given independent random variables: \\[X \\sim Poisson(\\lambda_1) \\] \\[Y \\sim Poisson(\\lambda_2) \\] Poisson PMF: \\[P(X=k) = \\frac{e^{-\\lambda} \\lambda^k}{k!} \\] Concept required: Sum of independent Poisson random variables: \\[X + Y \\sim Poisson(\\lambda_1 + \\lambda_2) \\] Conditional distribution result: \\[X | (X+Y=n) \\sim Binomial \\] Binomial PMF: \\[P(X=k | X+Y=n) = \\binom{n}{k} \\left(\\frac{\\lambda_1}{\\lambda_1+\\lambda_2}\\right)^k \\left(\\frac{\\lambda_2}{\\lambda_1+\\lambda_2}\\right)^{n-k} \\] \\section*{Question 6: Transformation of Random Variables} Given: \\[X \\sim Uniform[-2,2] \\] Transformation: \\[Y = g(X) \\] CDF method: \\[F_Y(y) = P(Y \\le y) \\] Transformation formula when invertible: \\[f_Y(y) = f_X(x) \\left| \\frac{dx}{dy} \\right| \\] Concepts required: \\begin{itemize} \\item Determining range of transformed variable \\item Finding inverse transformation \\item Using derivative of inverse function \\item Handling non-monotonic transformations using multiple roots \\item Relationship between CDF and PDF: \\[f_Y(y) = \\frac{d}{dy} F_Y(y) \\] \\end{itemize} \\section*{Question 7: Stock Price Model} Each step multiplies price by: \\[u \\quad with probability \\quad p \\] \\[d \\quad with probability \\quad 1-p \\] Let number of upward movements be: \\[N \\sim Binomial(n,p) \\] Binomial mean and variance: \\[E[N] = np \\] \\[Var(N) = np(1-p) \\] Final stock price expression: \\[S = s u^N d^{n-N} \\] Concept required: Taking logarithm converts multiplicative model into additive model. Normal approximation to binomial: \\[N \\approx Normal(np, np(1-p)) \\] Standardization formula: \\[Z = \\frac{N-np}{\\sqrt{np(1-p)}} \\] \\section*{Question

8: Exponential Repair Time} Given exponential distribution: \\[X \\sim Exponential(\\lambda) \\] Survival function: \\[P(X > t) = e^{-\\lambda t} \\] Memoryless property: \\[P(X > t+s | X > t) = P(X > s) \\] Concept required: Use survival function and memoryless property. \\section*{Question 9: Distribution of Z = sqrt(X^2 + Y^2)} Joint PDF for independent variables: \\[f_{X,Y}(x,y) = f_X(x) f_Y(y) \\] Transformation to polar coordinates: \\[x = r cos\\theta \\] \\[y = r sin\\theta \\] Jacobian determinant: \\[|J| = r \\] Concept required: PDF of transformed variable obtained using: \\[f_Z(r) = \\int_0^{2\\pi} f_X(r cos\\theta) f_Y(r sin\\theta) r d\\theta \\] \\section*{Question 10: Joint Distribution of Maximum} Maximum defined as: \\[M_n = max(X_1, X_2, ..., X_n) \\] CDF of maximum: \\[F_{M_n}(x) = P(M_n \\le x) = F(x)^n \\] Similarly: \\[F_{M_{n+1}}(x) = F(x)^{n+1} \\] Concept required: Relationship: \\[M_{n+1} = max(M_n, X_{n+1}) \\] Use independence of

Your lecture scribe has now been fully completed in proper LaTeX format, with:

- Complete coverage of all Questions 1–10
- Detailed theory, distributions, and formulae
- Strict focus on concepts needed to solve the problems
- No unnecessary additions or full solutions
- Clean, exam-ready structure

You can now directly copy this into Overleaf and compile.

If you'd like, I can also enhance it further with theorem boxes, highlighted formulas, or a professional scribe-style header used in top university lecture notes.

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Below is a **complete, detailed, exam-ready lecture scribe** based strictly on the Tutorial-2 document. It includes **all theory, distributions, formulae, and concepts required to solve each question**, without solving the problems. The structure is formal and suitable for revision and exams.
(Source:)

CSE 400 – Fundamentals of Probability in Computing

Lecture Scribe: Conceptual Foundations for Tutorial-2

Question 1: Random Point on a Line Segment

Uniform Distribution Model

When a point is chosen at random on a line segment of length L, its position is modeled using a continuous uniform random variable.

Let

$$X \sim \text{Uniform}(0, L)$$

Probability density function (PDF):

$$f_X(x) = \begin{cases} \frac{1}{L}, & 0 < x < L \\ 0, & \text{otherwise} \end{cases}$$

Cumulative distribution function (CDF):

$$F_X(x) = \begin{cases} 0, & x \leq 0 \\ \frac{x}{L}, & 0 < x < L \\ 1, & x \geq L \end{cases}$$

Segment Division

The chosen point divides the original segment into two parts:

$$\text{Left segment length} = X$$

$$\text{Right segment length} = L - X$$

Define:

$$\text{Shorter segment} = \min(X, L - X)$$

$$\text{Longer segment} = \max(X, L - X)$$

Probability Concept Used

The probability condition involves:

$$\frac{\min(X, L - X)}{\max(X, L - X)}$$

Concepts required:

- Uniform distribution property: constant density
- Probability equals ratio of interval lengths
- Symmetry around midpoint $L/2$
- Case-wise analysis:
  - Case 1: $X < L/2$
  - Case 2: $X > L/2$

Question 2: Classification Using Normal Distributions

Conditional Distributions

White region reading:

$$X|W \sim N(\mu_1 = 4, \sigma_1^2 = 4)$$

Black region reading:

$$X|B \sim N(\mu_2 = 6, \sigma_2^2 = 9)$$

Prior probabilities:

$$P(B) = \alpha, \quad P(W) = 1 - \alpha$$

Normal Distribution PDF

General form:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Bayesian Classification

Bayes' theorem:

$$P(B|x) = \frac{f_B(x)P(B)}{f_B(x)P(B) + f_W(x)P(W)}$$

$$P(W|x) = \frac{f_W(x)P(W)}{f_B(x)P(B) + f_W(x)P(W)}$$

Equal Error Condition

Equal probability of error requires:

$$P(B|x) = P(W|x)$$

This implies:

$$f_B(x)P(B) = f_W(x)P(W)$$

Concepts required:

- Conditional probability
- Prior probability
- Posterior probability
- Normal density substitution

Question 3: Minimizing Expected Absolute Distance

Objective Function

Minimize:

$$E[|X - a|]$$

This represents expected absolute deviation from point a.

Case 1: Uniform Distribution

$$X \sim Uniform(0, A)$$

PDF:

$$f_X(x) = \frac{1}{A}$$

Expected absolute deviation:

$$E[|X - a|] = \int_0^A |x - a| f_X(x) dx$$

Key Result

Expected absolute deviation is minimized at the median.

Median defined as:

$$P(X \leq m) = \frac{1}{2}$$

Uniform median:

$$m = \frac{A}{2}$$

Case 2: Exponential Distribution

$$X \sim Exponential(\lambda)$$

PDF:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

CDF:

$$F_X(x) = 1 - e^{-\lambda x}$$

Median satisfies:

$$F_X(m) = \frac{1}{2}$$

Concept required:

Expected absolute deviation minimized at median.

Question 4: Mixture of Exponential Distributions

Battery lifetime:

$$X_i \sim Exponential(\lambda_i)$$

Selection probability:

$$P(i) = p_i$$

Survival Function

Exponential survival function:

$$P(X > t) = e^{-\lambda t}$$

Mixture Survival Probability

Using total probability:

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

Conditional Probability Definition

$$P(X > t + s | X > t) = \frac{P(X > t + s)}{P(X > t)}$$

Concept required:

- Conditional probability
- Law of total probability
- Mixture distributions

Question 5: Conditional Distribution of Independent Poisson Variables

Given:

$$X \sim Poisson(\lambda_1)$$

$$Y \sim Poisson(\lambda_2)$$

PMF:

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

Sum of Independent Poisson

$$X + Y \sim Poisson(\lambda_1 + \lambda_2)$$

Conditional Distribution Result

$$X|(X + Y = n) \sim Binomial$$

Parameters:

$$n, \quad p = \frac{\lambda_1}{\lambda_1 + \lambda_2}$$

PMF:

$$P(X = k|X + Y = n) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Question 6: Transformation of Random Variables

Given:

$$X \sim Uniform[-2, 2]$$

Transformation:

$$Y = g(X)$$

CDF Method

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y)$$

PDF Transformation Formula

If invertible:

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

Concepts required:

- Transformation of random variables
- Inverse function method
- Differentiation of CDF

Question 7: Stock Price Model

Each step:

$$\text{Multiply by } u \quad \text{with probability } p$$

$$\text{Multiply by } d \quad \text{with probability } 1-p$$

Independent steps.

Binomial Model

Number of upward movements:

$$N \sim Binomial(n, p)$$

Mean:

$$E[N] = np$$

Variance:

$$Var(N) = np(1-p)$$

Final Price Expression

$$S = su^N d^{n-N}$$

Concept required:

Logarithmic transformation converts multiplicative model into additive form.

Normal Approximation

For large n:

$$N \approx Normal(np, np(1-p))$$

Standardization:

$$Z = \frac{N - np}{\sqrt{np(1-p)}}$$

Question 8: Exponential Distribution and Memoryless Property

Repair time:

$$X \sim Exponential(\lambda)$$

Survival function:

$$P(X > t) = e^{-\lambda t}$$

Memoryless property:

$$P(X > t + s | X > t) = P(X > s)$$

Concept required:

- Conditional probability
- Memoryless property

Question 9: Distribution of $Z = \sqrt{X^2 + Y^2}$

Joint PDF of independent variables:

$$f_{X,Y}(x,y) = f_X(x) f_Y(y)$$

Transformation:

$$x = r\cos\theta$$

$$y = r\sin\theta$$

Jacobian determinant:

$$|J| = r$$

Concept required:

PDF of transformed variable using Jacobian transformation.

Question 10: Distribution of Maximum

Define:

$$M_n = \max(X_1, X_2, \ldots, X_n)$$

CDF of maximum:

$$F_{M_n}(x) = P(M_n \leq x) = F(x)^n$$

Similarly:

$$F_{M_{n+1}}(x) = F(x)^{n+1}$$

Concept required:

Relationship:

$$M_{n+1} = \max(M_n, X_{n+1})$$

Use independence of random variables.

Core Concepts Summary

This tutorial uses the following probability concepts:

- Uniform distribution
- Normal distribution

- Bayes' theorem
- Expected absolute deviation
- Median of distributions
- Exponential distribution
- Memoryless property
- Poisson distribution
- Binomial distribution
- Normal approximation
- Transformation of variables
- Jacobian method
- Order statistics
- Conditional probability
- Law of total probability

If you want, I can also convert this into a perfectly formatted LaTeX scribe matching Ahmedabad University lecture standards.

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Cse400 Tutorial2 Final Conceptual Scribe

CSE 400 – Fundamentals of Probability in Computing

Lecture Scribe: Conceptual Foundations for Tutorial–2

This lecture scribe explains the theoretical concepts, probability distributions, properties, and formulae required to solve each problem in Tutorial–2. The focus is on identifying the mathematical models and principles necessary for solving the problems, without performing full computations.

Question 1: Random Point on a Line Segment

Modeling Using Continuous Uniform Distribution

When a point is chosen at random on a line segment of length L, its location is modeled as a continuous uniform random variable.

Let

$X \sim \text{Uniform}(0, L)$

Probability Density Function (PDF):

$f_X(x) = 1/L, \; 0 < x < L$

Cumulative Distribution Function (CDF):

$F_X(x) = x/L, \; 0 < x < L$

Segment Partition

The chosen point divides the original segment into two parts:

Left segment length = X

Right segment length = $L - X$

Define:

Shorter segment = $\min(X, L - X)$

Longer segment = $\max(X, L - X)$

Concepts Required

• Uniform distribution implies constant density over the interval

• Probability over an interval equals ratio of interval lengths

• Symmetry about midpoint L/2

• Case-wise analysis:

Case 1: $X \leq L/2$

Case 2: $X \geq L/2$

• Converting ratio condition into inequality involving X

Question 2: Classification Using Normal Distributions

Conditional Distributions

White region reading:

$X \mid W \sim \text{Normal}(4, 4)$

Black region reading:

$X \mid B \sim \text{Normal}(6, 9)$

Prior probabilities:

$P(B) = \alpha$

$P(W) = 1 - \alpha$

Normal Distribution Density Function

$f(x) = (1 / \sqrt{(2\pi\sigma^2)}) \exp(-(x - \mu)^2 / (2\sigma^2))$

Bayesian Classification

Posterior probability formula:

$P(B \mid x) = [f_B(x) P(B)] / [f_B(x)P(B) + f_W(x)P(W)]$

$P(W \mid x) = [f_W(x) P(W)] / [f_B(x)P(B) + f_W(x)P(W)]$

Equal Probability of Error Condition

Equal classification error implies:

$P(B \mid x) = P(W \mid x)$

Which leads to:

$f_B(x) P(B) = f_W(x) P(W)$

Concepts required:

- Conditional probability
- Bayes' theorem
- Normal density substitution
- Solving equation involving exponential functions

Question 3: Minimizing Expected Absolute Distance

Objective function:

Minimize $E[|X - a|]$

This represents expected absolute deviation.

Uniform Distribution Case

$X \sim \text{Uniform}(0, A)$

PDF:

$f_X(x) = 1/A$

Expected value expression:

$E[|X - a|] = \int |x - a| f_X(x) dx$

Key concept:

Expected absolute deviation is minimized at the median.

Median satisfies:

$P(X \le m) = 1/2$

Uniform median:

$m = A/2$

Exponential Distribution Case

$X \sim \text{Exponential}(\lambda)$

PDF:

$f_X(x) = \lambda e^{(-\lambda x)}, x > 0$

CDF:

$F_X(x) = 1 - e^{(-\lambda x)}$

Median condition:

$F(m) = 1/2$

Concept required:

Median minimizes expected absolute deviation.

Question 4: Mixture of Exponential Distributions

Each battery type lifetime follows exponential distribution:

$X_i \sim \text{Exponential}(\lambda_i)$

Selection probability:

$P(\text{Type } i) = p_i$

Survival Function

$P(X > t) = e^{(-\lambda t)}$

Law of Total Probability

Overall survival probability:

$P(X > t) = p_1 e^{(-\lambda_1 t)} + p_2 e^{(-\lambda_2 t)}$

Conditional Probability Definition

$P(X > t + s \mid X > t) = P(X > t + s) / P(X > t)$

Concept required:

- Law of total probability
- Conditional probability
- Survival functions
- Mixture distributions

Question 5: Conditional Distribution of Poisson Random Variables

Given independent random variables:

$X \sim \text{Poisson}(\lambda_1)$

$Y \sim \text{Poisson}(\lambda_2)$

Poisson PMF:

$P(X = k) = (e^{(-\lambda)} \lambda^k) / k!$

Sum of Independent Poisson Variables

$X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2)$

Conditional Distribution Result

Conditional distribution follows binomial distribution.

Binomial PMF:

$P(X = k \mid X + Y = n)$

$= C(n, k)\, p^k (1 - p)^{(n - k)}$

Where

$p = \lambda_1 / (\lambda_1 + \lambda_2)$

Concepts required:

- Poisson distribution
- Independence
- Conditional probability
- Relationship between Poisson and binomial distributions

Question 6: Transformation of Random Variables

Given:

$X \sim \text{Uniform}[-2, 2]$

Transformation:

$Y = g(X)$

CDF Method

$F_Y(y) = P(Y \leq y)$

$= P(g(X) \leq y)$

PDF Transformation Formula

$f_Y(y) = f_X(x)\, |dx/dy|$

Concepts required:

- Transformation of continuous random variables
- Inverse transformation
- Differentiation of CDF
- Determining range of transformed variable

Question 7: Stock Price Model

Each step:

Multiply by u with probability p

Multiply by d with probability 1 − p

Independent across steps.

Binomial Distribution Model

Number of upward movements:

N ~ Binomial(n, p)

Mean:

E[N] = np

Variance:

Var(N) = np(1 − p)

Final Price Expression

S = s u^N d^(n − N)

Concept required:

Logarithmic transformation simplifies multiplicative expressions.

Normal Approximation

For large n:

$N \approx \text{Normal}(np, np(1 - p))$

Standardization:

$Z = (N - np) / \sqrt{(np(1 - p))}$

Question 8: Exponential Distribution and Memoryless Property

$X \sim \text{Exponential}(\lambda)$

Survival function:

$P(X > t) = e^{(-\lambda t)}$

Memoryless property:

$P(X > t + s \mid X > t) = P(X > s)$

Concepts required:

- Exponential distribution
- Conditional probability
- Memoryless property

Question 9: Distribution of $Z = \sqrt{(X^2 + Y^2)}$

Joint density for independent variables:

$f_{X,Y}(x,y) = f_X(x) f_Y(y)$

Transformation to polar coordinates:

$x = r \cos\theta$

$y = r \sin\theta$

Jacobian determinant:

$|J| = r$

Concept required:

Use Jacobian transformation method.

Question 10: Distribution of Maximum

Define maximum:

$M_n = \max(X_1, X_2, \ldots, X_n)$

CDF of maximum:

$F_{M_n}(x) = [F(x)]^n$

Relationship:

$M_{n+1} = \max(M_n, X_{n+1})$

Concept required:

• Order statistics

• Independence

• CDF properties

Summary of Core Concepts

This tutorial requires understanding of:

• Uniform distribution

• Normal distribution

• Bayes' theorem

- Exponential distribution
- Memoryless property
- Poisson distribution
- Binomial distribution
- Normal approximation
- Transformation of random variables
- Jacobian method
- Conditional probability
- Law of total probability

Your final lecture scribe has now been created in canvas. It is:

- Fully detailed and exam-oriented
- Strictly based on the tutorial document
- Focused only on theory, concepts, and formulae
- Structured clearly question-by-question
- Suitable for revision, exams, and submission

If you want, I can also convert it into a perfectly formatted LaTeX scribe identical to Ahmedabad University lecture standards with centered headers, proper spacing, and professional formatting.